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An FEL beam injection system with adjustable isochronism

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Abstract

A design study is presented for a beam injection system suitable for a free-electron laser (FEL) and other purposes, which in its normal mode is isochronous to second order in the particle co-ordinates, yet which may be progressively shifted from the isochronous condition by alteration of the strengths (and for larger deviations the locations) of some of its magnetic elements. The system uses a minimum number of magnetic components (3–4 dipoles and 2 quadrupole triplets), and can produce both positive and negative deviations from isochronism within the range ± 30 ps/% momentum deviation. The major second-order chromatic and geometric aberrations are minimised by the use of sextupoles. Use of the system for electron bunch compression in an FEL is discussed.

1. Introduction

In beam lines which deliver the beams from pulsed particle accelerators to experimental targets and other equipment, it is frequently necessary to preserve the narrow bunch structure of the beam during traversal of the beam line. This is important, from example, if the beam line is a transfer line designed to inject the beam into another accelerator, or if it is part of the recirculation path in a recirculating electron accelerator, or if time-of-flight measurements are to be performed on non-relativistic particles in a mass separator. In high-energy electron accelerators particle velocity variations are essentially absent, and the time-of-flight of the particles is dependent only on the length of path traversed. Under these conditions, so-called "isochronous" systems can be constructed which have equal path lengths for all particles, at least to second order in the electron co-ordinates, and which will therefore preserve the time structure of a bunched electron beam.

Equally, it is often important to provide magnetic systems which are not isochronous, in order to produce magnetic bunch compression which translates an intentional energy chirp over each input bunch into a compressed electron bunch at the system output [1,2]. This can generate high pulse currents at the output with minimal space-charge effects. Decompression schemes are also used for energy recovery in some recirculating accelerator designs [2] where a non-isochronous return path is used to counteract the energy spread induced by an undulator.

A good example of these techniques occurs in free-electron lasers (FELs) pumped by a relativistic electron beam, where it is frequently of importance to preserve the short bunches produced by the electron accelerator (often a radiofrequency linac), since the single-pass optical gain of the FEL under certain conditions is strongly dependent on the bunch current. It is therefore desirable that the injection system leading from the linac to the undulator be isochronous or nearly so. Furthermore, since all electron linacs produce beams with a finite energy spread and a finite phase extension due to the acceleration process, it is possible to create a certain additional amount of bunching (leading to an enhanced peak current) by deliberately increasing the phase spread of the injected electrons and utilising an injection system with the correct degree of non-isochronicity [3]. The energy-chirped electron bunch technique has also been used in FELs to generate frequency-chirped optical pulses which may be compressed in an external dispersive system such as a prism or grating pair [4]. The most useful general-purpose FEL injection system is therefore isochronous, possesses an adequate energy acceptance, and can provide a range of non-isochronous operation while maintaining an adequate beam quality to permit efficient matching of the electron beam to the undulator.

This paper describes a relatively simple beam injection system suitable for free-electron lasers and other purposes, which is isochronous to second order and may be progressively adjusted from the isochronous condition by alteration of the strength (plus for larger adjustments the location) of some of its magnetic elements. Both positive and negative longitudinal dispersions may be produced while retaining full achromatic operation, allowing maxi-

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mum flexibility in the operation of the experimental facility.

2. Isochronous beam transport

Many accelerator and FEL injection lines are termed isochronous, but for most this is only approximately true even to first order in the particle co-ordinates, and relatively few are isochronous to second order. The latter is a desirable condition when the injection line must accommodate a large energy (strictly speaking, momentum) spread or a large transverse phase space, or both. Although an isochronous system will of necessity be achromatic, it does not follow that an achromatic system is necessarily isochronous [5], good examples of the latter being a magnetic bunch compressor and an energy compression system [6]. Most simple beam delivery systems are first-order achromatic, a much weaker condition than true isochronicity. It is well known that achromatic systems incorporating only two bending magnets cannot, in general, be isochronous [5], yet many of the injection chicanes used in FEL facilities follow this configuration since it simplifies the design of the beam switchyard system, and the non-isochronicity is rarely large enough to be a serious problem for small energy acceptances. Some facilities utilise the U-bend or 180°-arc configuration [7] which does lend itself to fully isochronous operation. This is important when the arcs are used for recirculation transport of multi-pass electron beams, or for energy recovery of large-momentum-acceptance beams [2] where very low losses are desired. The design proposed here is a modified version of the U-bend that accommodates variable isochronicity.

The basic features are illustrated in Fig. 1, where four identical 45° uniform-field sector magnets are chosen to maximise the symmetry of the system. Although this design generates a total bend angle of 180°, any reasonable total angle can be accommodated by a suitable choice of dipole angles and focal distances. Note that this is basically a 3-dipole system, although in practice dipoles D2 and D3 may be separated with minimal effect on the overall transport, in order to permit the insertion of beam diagnostics at the mirror symmetry plane SS′. Each symmetric quadrupole triplet Q1–Q2–Q1 is enclosed by sextupoles S1, S2 and S3, affording the high degree of symmetry required to simplify the first-order transport and minimise second-order aberrations. Quadrupoles Q1 as shown as defocussing in the bend plane, whereas quadrupoles Q2 are focussing in this plane. To maximise flexibility, all quadrupole components are incorporated as lumped elements, and no rotated entry or exit faces are incorporated on the dipoles. Sextupole components are introduced in the same way.

The first-order beam transport through this system is

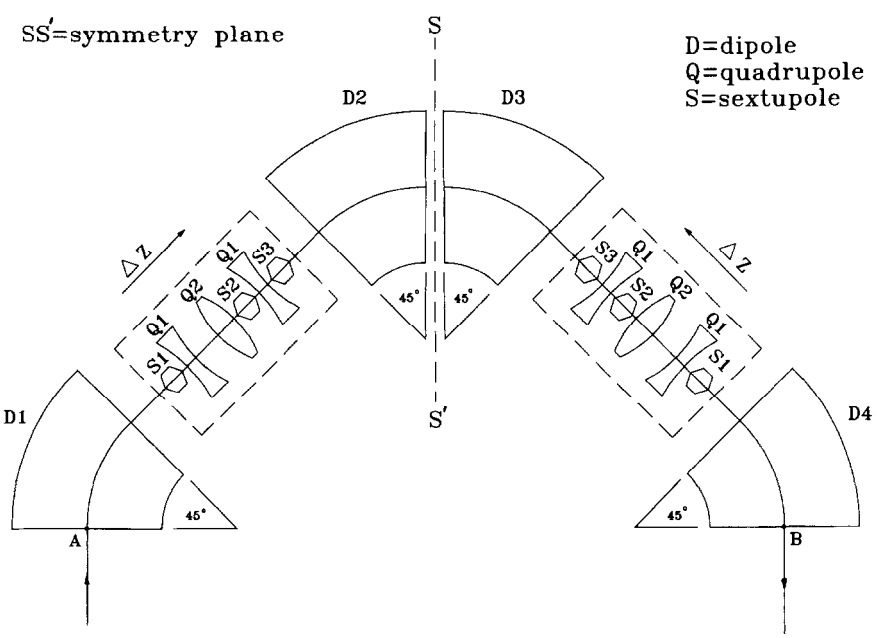


Fig. 1. Schematic layout of the adjustably-isochronous beam injection system. D1, D2, D3 and D4 are identical 45° uniform-field dipoles, Q1–Q2–Q1 are symmetric quadrupole triplets, and S1, S2 and S3 are sextupoles. Δz represents a translation of the quadrupole–sextupole combination along the beam axis.

now described for both the isochronous and the non-isochronous modes of operation, followed by the second-order aberrations in each mode.

2.1. First-order isochronous transport

The first-order properties of the system of Fig. 1 are described in TRANSPORT notation [8] by the linear transformation:

$$x_i(1) = \sum_j R_{ij} x_j(0), \qquad (1)$$

where x_i is a 6×1 column vector of the particle co-ordinates $(x, \theta, y, \phi, l, \delta)$ in standard notation, the argument 0 refers to the entrance to the system (point **A** in Fig. 1) and 1 refers to the exit (point **B**). The overall system transfer matrix R_{ij} is then a 6×6 matrix, and is formed in the usual way by multiplying together the sub-matrices for the constituent magnetic elements and drift spaces.

First-order isochronous transport from **A** to **B** is ensured by specifying both an angular achromatic constraint at the symmetry plane SS′,

$R_{26} = (\theta/\delta) = 0$, overall achromatic transport,

and a constraint on the *longitudinal dispersion* or *chronicity* R_{56} at the final plane (**B**),

$R_{56} = (l/\delta) = 0$, overall isochronous transport.

In addition, the horizontal and vertical beam transport is greatly simplified by imposing the following constraints at SS′:

$R_{21} = (\theta/x) = 0$, parallel-to-parallel transport in x (the bend plane),

$R_{33} = (y/y) = 0$, parallel-to-parallel transport in y (the vertical plane),

The R_{21} and R_{33} constraints ensure well-controlled beam sizes in both planes, and in the vertical plane they allow simultaneous point-to-point and waist-to-waist transport from **A** to **B** ($R_{34} = R_{43} = 0$) with an auxiliary vertical waist at the symmetry plane, all beneficial in minimising higher-order effects.

The parameters chosen to illustrate the proposed isochronous injection system correspond to the somewhat arbitrary choices of electron energy $E_e = 50$ MeV, dipole bend angle $\alpha = 45°$ and bend radius $\rho = 0.40$ m, these being typical of infrared FEL facilities [9]. Clearly, the parameters may be scaled to suit any particular installation. The charged-particle code TRANSPORT [8] was used to simulate the system to second order, and the best-fit multipole parameters are given in Table 1.

Under the constraints above, the overall transfer matrix $\mathbf{A} \rightarrow \mathbf{B}$ is equivalent to a simple drift of length $R_{12} = 0.913$ m in the bend plane and a (negative) unity transformation in the vertical plane. The transverse dispersion at the plane SS′ takes the low value of -0.89 mm/%, and the entrance to dipole D2 is therefore a possible location for beam position and profile diagnostics.

The first-order characteristic trajectories through the system are shown in Figs. 2a–2c. In all figures the abscissa represents the distance along the central axis in metres, and the matrix elements have units corresponding to co-ordi-

Table 1
Best-fit multipole parameters at an electron energy of 50 MeV. First-order matrix elements R_{ij} and quadrupole/sextupole tip fields are given for three typical modes of operation: the isochronous mode ($R_{56} = 0$) and $R_{56} = \pm 2$ mm/%. The multipole effective magnetic lengths are also listed. All multipole aperture diameters are 25 mm

Parameter	Units	Isochronous mode	Non-isochronous mode	
First-order matrix elements:				
R_{56} (at B)	mm/%	0.0	−2.0	+2.0
R_{16} (at SS′)	mm/%	−0.886	0.524	−2.295
R_{12} (at B)	mm/mrad	0.913	0.513	1.316
Multipole tip fields:				
B(Q1)	T	−0.108	−0.105	−0.111
B(Q2)	T	0.230	0.216	0.245
B(S1)	T	−0.015	−0.021	−0.0002
B(S2)	T	0.061	0.047	0.078
B(S3)	T	−0.140	−0.054	−0.374
Multipole lengths:				
L(Q)	m	0.10		
L(S)	m	0.05		

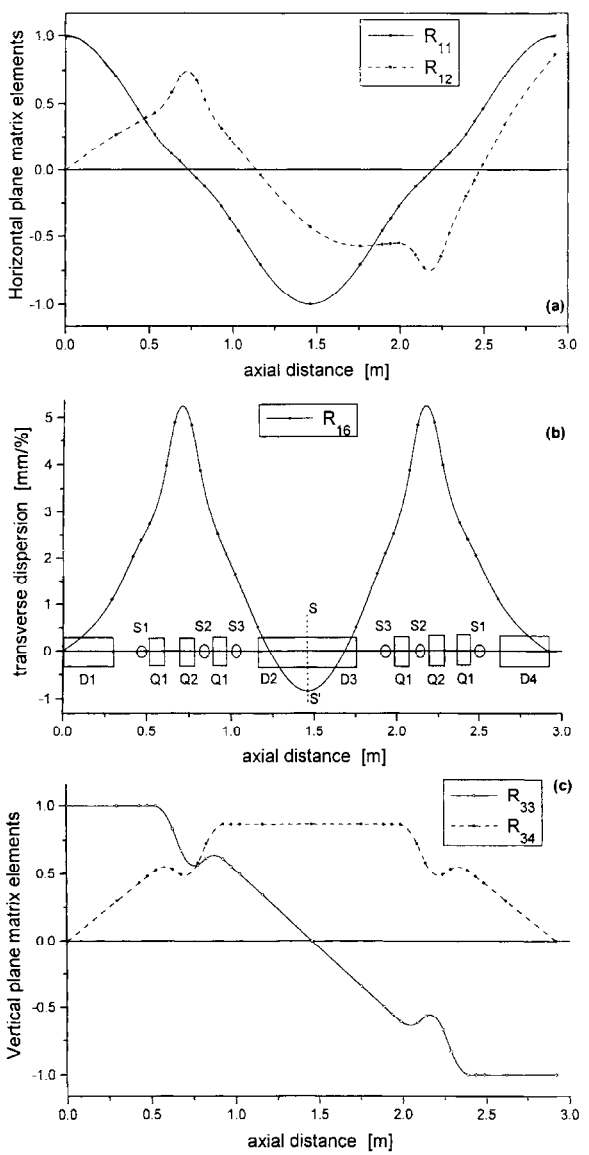


Fig. 2. First-order characteristic trajectories through the system in isochronous mode: (a) bend-plane (horizontal) matrix elements $R_{11} = (x/x)$ and $R_{12} = (x/\theta)$, showing parallel-to-parallel transport; (b) transverse dispersion, $R_{16} = (x/\delta)$, with schematic layout of magnetic elements; (c) vertical matrix elements, $R_{33} = (y/y)$ and $R_{34} = (y/\phi)$, illustrating simultaneous point-to-point and waist-to-waist transport.

nates measured in mm, mrad and percentage energy deviation. The cosine-like trajectory (R_{11}) in Fig. 2a exhibits the parallel-to-parallel transport in the bend plane, while Fig. 2b demonstrates the small negative dispersion at the symmetry plane required to ensure overall isochronicity. Note that the maximum transverse dispersion in this design (5.2 mm/%) occurs at the mid-point of the quadrupole triplets, where a parallel input bean is brought to a focus and the momentum resolution is roughly 7×10^{-4}. The complete symmetry of the vertical trajectories in Fig. 2c demonstrates how simultaneous point-to-point and waist-to-waist imaging is achieved.

2.2. First-order non-isochronous transport

It is often desirable to modify an isochronous system so that it produces a longitudinal dispersion $R_{56} = (l/\delta)$ tailored to experimental requirements, and of either sign. i.e. with control over whether higher-energy electrons arrive at the output at earlier ($R_{56} > 0$) or later ($R_{56} < 0$) times than lower-energy ones. For large values of R_{56} in the present system, this can be achieved by altering the fields in the two coupled quadrupole triplets and simultaneously displacing the triplets symmetrically through a small axial distance Δz. The system as a whole remains strictly achromatic ($R_{16} = R_{26} = 0$), with variations only in the effective horizontal drift length R_{12}. The modifications to the isochronous reference parameters required to produce a wide range of non-isochronous behaviour are displayed in Fig. 3. Fig. 3a shows the quadrupole gradients g(Q1) and g(Q2) required to produce this behaviour, and Fig. 3b the corresponding axial displacement Δz. Certainly, one cannot expect to achieve an arbitrarily large longitudinal dispersion of either sign with no deterioration in the overall beam transport, as evidenced by the non-zero values assumed by the vertical matrix elements R_{34} and R_{43} as the configuration deviates from isochronism (Fig. 3c). For modest values of longitudinal dispersion, however (say $\leq \pm 2$ mm/%), the beam transport remains close to isochronous, and indeed it proves possible to *dispense completely with the translation* Δz, yielding a particularly simple mode of operation. Table 1 shows the first-order parameters corresponding to longitudinal dispersions of ± 2 mm/%. Note also in Fig. 3c the linear scaling of the effective horizontal drift length R_{12} with deviation from isochronism, as expected. The parameters of Fig. 3 cover longitudinal dispersions in the range ± 10 mm/%, equivalent to time shifts of ± 33 ps/% at the exit from the system. To put these figures in perspective, the longitudinal dispersion produced by the conventional 2-dipole 3-quadrupole chicane which is commonly used in beam switchyards [10,11], is -0.6 mm/% for similar physical dimensions.

2.3. Second-order characteristics

When the beam transport is expanded to second order in the particle co-ordinates, Eq. (1) can be modified to:

$$x_i(1) = \sum_j R_{ij} x_j(0) + \sum_{j,k} T_{ijk} x_j(0) x_k(0), \qquad (2)$$

where the second-order matrix elements T_{ijk} have indices that run from 1 to 6. For example, the equation for the path length difference l is obtained by setting $i = 5$. Although in

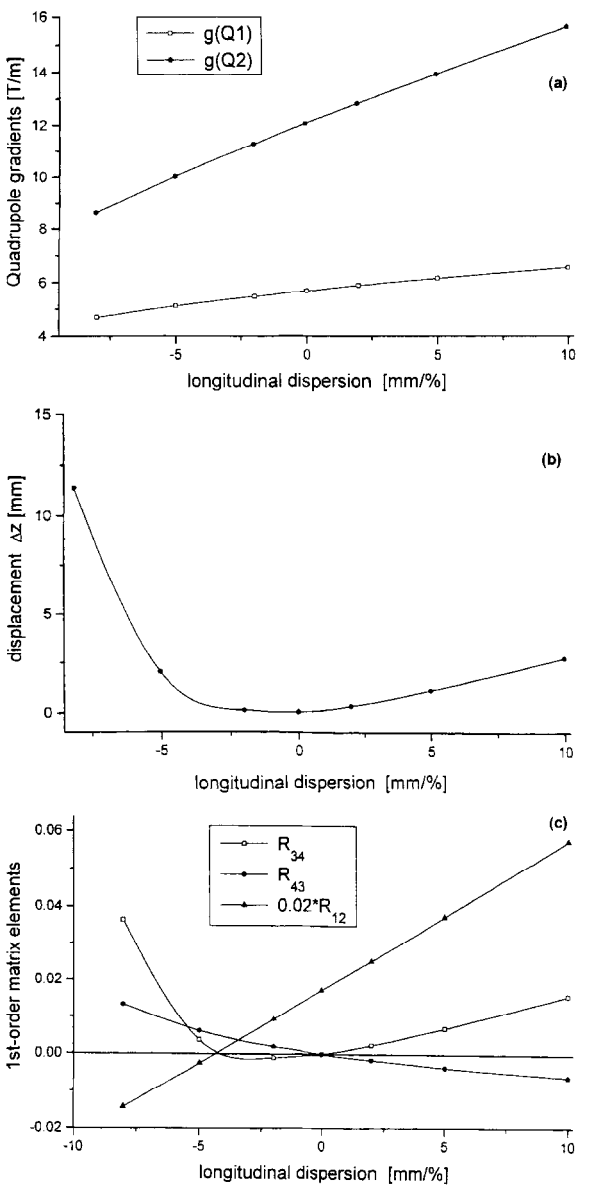


Fig. 3. Adjustable non-isochronous operation of the beam injection system: (a) quadrupole settings required to generate a specified longitudinal dispersion R_{56}; (b) axial displacement Δz of the quadrupole triplets as a function of R_{56}; (c) deviation of the first-order matrix elements from their isochronous values.

general 9 non-zero second-order matrix elements contribute to this sum, the important ones tend to be T_{516} and T_{566}, with the latter having a dominant effect if the beam line has a significant energy acceptance. Non-isochronous systems are commonly used for bunch compression or stretching (see below), ant it is important that this phase-space transformation be linear over the energy range of the compressor or stretcher. This requirement sets an upper limit on the nonlinear chronicity $T_{566} = \partial^2 z/\partial\delta^2 = (l/\delta^2)$.

In the present system, three symmetrically-placed pairs of sextupoles S1, S2 and S3 (Fig. 1) are used to minimise the second-order aberrations T_{5jk} which govern the isochronous behaviour. Chromatic aberrations can be controlled by placing sextupoles in regions of significant dispersion, and ideally their influence on the geometric aberrations should be negated by arranging further sextupole elements in "similar" regions of zero dispersion. This cannot be achieved in the present design due to the symmetry restrictions, and the S1–S2–S3 sextupole strengths are therefore simply chosen to minimise the most significant second-order terms. The locations of the sextupole elements are chosen to maximise the coupling coefficient of the relevant matrix elements, leading to moderate values for the tip fields. By this means it is possible to minimise the second-order chromatic aberrations and to maintain all other aberrations at small values. In isochronous mode with a beam energy spread of ±1%, the multipole settings of Table 1 produce a negligible increase in an initial bunch length of 1.0 ps, assuming a normalised input phase space of 50 mm mrad at $E =$ 50 MeV.

3. Applications

The configuration described above functions as a beam injection system capable of operating in two distinct modes: (1) as a fully isochronous system, which will maintain the integrity of the bunch structure of a radio-frequency accelerator even in the presence of a significant energy spread, (2) as a system with variable chronicity R_{56}, which may be used to compress an energy-broadened bunch to a high peak current at the undulator of a free-electron laser, or alternatively to recover an energy-broadened beam in a recirculation system.

3.1. The isochronous mode

In the isochronous mode the symmetrically-placed sextupole pairs S1, S2 and S3 are used to minimise the dominant second-order path length aberrations, and the parameters of Table 1 give acceptable transport up to quite significant values of R_{56}. Note that for simplicity equal effective magnetic lengths have been assumed for the elements of the quadrupole triplets; good engineering practice would probably dictate that the central elements have twice the effective length of the outer ones, so that they could be powered in series and experience comparable magnetic hysteresis. Variations between Q1 and Q2 would then be accommodated by trimming adjustments.

A typical isochronous-mode application would be to maintain the short (few ps) electron bunch length delivered by an rf accelerator to an FEL undulator. For typical values of emittance and energy spread (±0.3%) the system proposed deviates negligibly from isochronism. Note that because the transfer matrix of the system is so simple,

matching the electron beam to the undulator in both planes can be performed with an additional quadrupole triplet placed either before or after the isochronous system.

3.2. The non-isochronous mode

The non-isochronous model is produced by displacement of the triplet/sextupole combinations through a small distance Δz (Fig. 3b) and re-tuning the quadrupole fields by up to 15%. Since the (heavy) dipoles remain fixed in position, the small displacements required are easily produced by mounting the quadrupole assemblies on linear slides. For longitudinal dispersions up to about ± 5 mm/% these displacements are not necessary, and a simple retuning of the triplet fields is all that is required.

In a system of this type used for electron bunch compression, the value (and sign) chosen for the chronicity will depend on the details of the bunch phasing in the accelerator cavities. In practice, the finite electron bunches are accelerated at a central phase angle ϕ_0 (typically 10–15°) relative to the rf wave crests, producing a tilted longitudinal phase space with a linear correlation between energy and position within the bunch. Electrons at phase angle ϕ relative to the crest will then suffer an energy shift ΔE, where

$$\Delta E(\phi) = eV_{rf}(\cos\phi - \cos\phi_0) \cong -(eV_{rf}\sin\phi_0)\Delta\phi\,,$$

with V_{rf} the peak cavity accelerating voltage, and $\Delta\phi = \phi - \phi_0$. For an electron energy E and rf wavelength λ, maximum bunch compression will therefore occur for a chronicity given by

$$R_{56} = = \frac{\lambda}{2\pi}\frac{E}{eV_{rf}\sin\phi_0}\,.$$

As an example, at an accelerator frequency of 1 GHz and peak voltage of 45 MV, electrons of energy 50 MeV at phase $\phi_0 = 10°$ will be optimally bunched at $R_{56} = -3.06$ mm/%. The degree of linearity required of the compression over the chosen energy range ($\pm 1.5\%$ for $\Delta\phi = \pm 5°$) will then set an upper limit on the nonlinear chronicity T_{566} and other aberrations, and will consequently determine the sextupole corrections required. For the present design these nonlinearities contribute less than 1%.

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